

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Cybersecurity

Time: 1.5 Hours

Semester / Programme: 7th Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

- 1(a)** Why is the classic model for network security essential before designing a secure system? [1]
- 1(b)** Differentiate briefly between Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPsec. [1]
- 1(c)** Why is DNS poisoning considered more damaging than ARP poisoning? [1]
- 1(d)** What is the primary function of a Demilitarized Zone (DMZ) in network architecture? [1]
- 1(e)** Distinguish between a corporate Policy and a Law in the context of Cyber Law and Ethics. [1]

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

- Q.2.** (a) Explain the OSI/TCP-IP-based Model for Network Access Security. How does it handle unauthorized access threats at different layers? [3]
- (b) Discuss the implementation layers for security protocols within the TCP/IP protocol stack and analyze their security implications. [2]
- Q.3.** (a) Explain the operational mechanics of Internet Key Exchange (IKE) in IPsec. Differentiate between its two phases. [3]
- (b) What is a Man-in-the-Middle (MitM) attack? Illustrate how a replay attack differs from a standard MitM eavesdropping scenario. [2]
- Q.4.** (a) Explain how ARP Poisoning works step-by-step. What countermeasures can be deployed to mitigate this vulnerability on a local network? [3]
- (b) Compare Network Intrusion Detection Systems (NIDS) and Host-based Intrusion Prevention Systems (HIPS) in terms of placement and visibility. [2]
- Q.5.** (a) Classify the major types of Law relevant to cybersecurity and computer crimes. Provide a brief description of each. [2.5]
- (b) Discuss the scope and key protections offered by U.S. Copyright Law in the digital domain, particularly regarding software and digital media. [2.5]

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